

Supplementary Material

This supplement gives the complete edge lists referred to in Sec. 4.5 of the manuscript and a per-edge scan of both control graphs. Both graphs are reproducible on a current install: `nx.random_labeled_tree(15, seed=42)` and `nx.gnm_random_graph(15, 27, seed=42)` return exactly the edge lists below, verified on NetworkX 3.6.1 against the authors' recorded data. The depths in S3 and S4 are computed from the projection average over the degenerate ground-state manifold. Where that manifold is more than one-dimensional the value differs from one taken from a single returned eigenvector, so these numbers are not in general the ones a single-state scan reports.

S1. Tree, $N = 15$, 14 edges

(0, 7)	(0, 11)	(1, 6)	(1, 10)	(1, 12)	(2, 3)	(2, 11)
(3, 4)	(3, 9)	(4, 8)	(5, 10)	(10, 11)	(11, 14)	(13, 14)

S2. Random graph, $N = 15$, 27 edges

(0, 1)	(0, 6)	(0, 8)	(0, 11)	(0, 12)	(1, 6)	(1, 9)
(1, 10)	(1, 11)	(2, 3)	(2, 12)	(3, 4)	(3, 7)	(3, 11)
(4, 5)	(4, 9)	(4, 12)	(5, 9)	(5, 12)	(5, 13)	(6, 8)
(6, 11)	(7, 8)	(8, 9)	(8, 14)	(10, 11)	(12, 13)	

S3. Per-edge scan, tree

Edge	Valley s	Depth	Eligible
0-7	0.1	0.013914957850685944	True
0-11	1.7000000000000002	0.06072740591968412	True
1-6	0.9500000000000001	0.055888671836528125	True
1-10	1.9000000000000001	0.058909125044506294	True
1-12	0.9500000000000001	0.05588867183652885	True
2-3	1.7000000000000002	0.07497728126356876	True
2-11	0.30000000000000004	0.03516767284711647	True
3-4	1.9500000000000002	0.06969080148864445	True
3-9	0.25	0.0428899418460891	True
4-8	0.05	0.007249731524320929	True
5-10	0.15000000000000002	0.010517228514336535	True
10-11	1.6	0.05097551083961485	True
11-14	1.7000000000000002	0.06072740591968409	True
13-14	0.1	0.01391495785068425	True

S4. Per-edge scan, random graph

Edge	Valley s	Depth	Eligible
0-1	—	—	False
0-6	1.4000000000000001	0.007318155095876688	True
0-8	0.65	0.0030782749921849373	True

0-11	—	—	False
0-12	1.4500000000000002	0.028256406302427983	True
1-6	0.5	0.0021837856387860732	True
1-9	1.2000000000000002	0.022008043117070775	True
1-10	0.65	0.002755309907965048	True
1-11	1.9500000000000002	0.014813179770286788	True
2-3	0.75	0.007699733490266258	True
2-12	0.8500000000000001	0.013123120267232141	True
3-4	0.9500000000000001	0.013408814967931282	True
3-7	1.1	0.014497177684680673	True
3-11	—	—	False
4-5	1.0	0.020282366213569797	True
4-9	1.0	0.001804772360859741	True
4-12	0.5	0.002619950758697781	True
5-9	1.1500000000000001	0.03281352928974185	True
5-12	1.35	0.008463051418390988	True
5-13	0.7000000000000001	0.04552438308690665	True
6-8	0.9	0.005550999941196011	True
6-11	1.05	-0.0014632737429298992	True
7-8	1.2000000000000002	0.06415343152003936	True
8-9	0.65	0.0050248082404896754	True
8-14	1.2000000000000002	0.050837261219369934	True
10-11	0.9500000000000001	0.009828226256085787	True
12-13	1.1	0.02323082712861807	True
